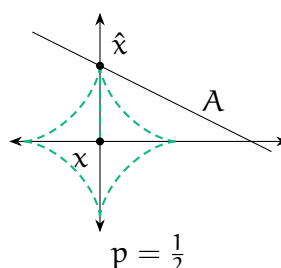
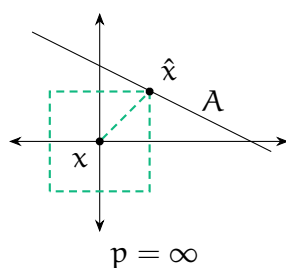
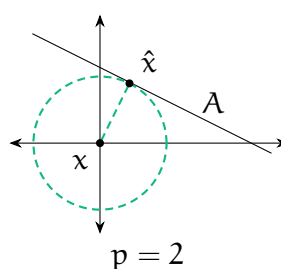
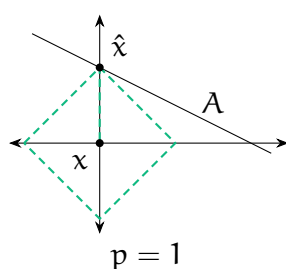


FUNCTIONAL ANALYSIS

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CHAPTER 1

Hilbert space



1.1 Inner product spaces

1.1.1. Definition (Inner product). — Let V be a $\mathbb{C}$ -vector space. An **inner product** on V is a map

$$\langle \cdot, \cdot \rangle : V \times V \longrightarrow \mathbb{C} \\ (x, y) \longmapsto \langle x, y \rangle$$

satisfying the following properties:

1. $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle \quad \forall x, y, z \in V$ (additivity in first component).
2. $\langle \lambda x, y \rangle = \lambda \langle x, y \rangle \quad \forall \lambda \in \mathbb{C}, x, y \in V$ (homogeneity or linearity in first component).
3. $\langle x, y \rangle = \overline{\langle y, x \rangle} \quad \forall x, y \in V$ (conjugate symmetry).
4. $\langle x, x \rangle \geq 0 \quad \forall x \in V$ (positivity).
In particular, $\langle x, x \rangle = 0 \iff x = 0 \in V$ (definiteness).

We then say $\langle \cdot, \cdot \rangle$ is an inner product defined on V , making it into an **inner product space**.

The number $\|x\| = \sqrt{\langle x, x \rangle}$ is called the **norm** of x &

$$d(x, y) = \|x - y\| = \sqrt{\langle x - y, x - y \rangle}$$

makes V a metric space (V, d) . Every inner product space is a metric space, but the converse is not true.

1.1.2. Example. — Let $V = \mathbb{F}^n$ ($\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$) and $x, y \in \mathbb{F}^n : x = (x_1, \dots, x_n), y = (y_1, \dots, y_n)$. Then

$$\langle x, y \rangle = x_1 \bar{y}_1 + \dots + x_n \bar{y}_n \quad (1.1.1)$$

is the usual, standard or Euclidean inner product on $\mathbb{F}^n$. For $\lambda_1, \dots, \lambda_n > 0, \lambda_1, \dots, \lambda_n \in \mathbb{F}$,

$$\langle x, y \rangle = \lambda_1 x_1 \bar{y}_1 + \dots + \lambda_n x_n \bar{y}_n \quad (1.1.2)$$

is also an inner product on $\mathbb{F}^n$.

1.1.3. Example. — Let $V = C[a, b]$ be the $\mathbb{R}$ -vector space of all continuous functions $f \in \mathbb{R}^{[a, b]}$ ($-\infty < a < b < \infty$). Then an inner product on V is given by

$$\langle f, g \rangle = \int_a^b f(x)g(x)dx. \quad (1.1.3)$$

In particular, if $V = C[-1, 1]$ then

$$\langle f, g \rangle = \int_{-1}^1 f(x)g(x)dx \quad (1.1.4)$$

and if $V = C[0, 1]$ then

$$\langle f, g \rangle = \int_0^1 f(x)g(x)dx. \quad (1.1.5)$$

If $f \in C^{[a,b]}$ instead, then

$$\langle f, g \rangle = \int_a^b f(x)\overline{g(x)}dx. \quad (1.1.6)$$

1.1.4. Theorem (Basic properties of inner product). — *Let V be an $\mathbb{F}$ -inner product space. Then the following hold:*

1. For all $y \in V$ (fixed), there is a linear functional $V \rightarrow \mathbb{F}$ defined by $x \mapsto \langle x, y \rangle$.
2. $\langle x, 0 \rangle = \langle 0, x \rangle = 0 \quad \forall x \in V$.
3. $\langle x, y + z \rangle = \langle x, y \rangle + \langle x, z \rangle \quad \forall x, y, z \in V$.
4. $\langle x, \lambda y \rangle = \bar{\lambda} \langle x, y \rangle \quad \forall \lambda \in \mathbb{F}, x, y \in V$.

Proof. 1. Let $\lambda \in \mathbb{F}, a, b \in V$. Then

$$\lambda a + b \mapsto \langle \lambda a + b, y \rangle = \langle \lambda a, y \rangle + \langle b, y \rangle = \lambda \langle a, y \rangle + \langle b, y \rangle.$$

2. Observe that

$$\langle 0, x \rangle = \langle x, x \rangle + \langle -x, x \rangle = \langle x, x \rangle - \langle x, x \rangle = 0.$$

$$\text{Hence, } \langle x, 0 \rangle = \overline{\langle 0, x \rangle} = \bar{0} = 0.$$

$$3. \langle x, y + z \rangle = \overline{\langle y + z, x \rangle} = \overline{\langle y, x \rangle + \langle z, x \rangle} = \overline{\langle y, x \rangle} + \overline{\langle z, x \rangle} = \langle x, y \rangle + \langle x, z \rangle.$$

$$4. \langle x, \lambda y \rangle = \overline{\langle \lambda y, x \rangle} = \bar{\lambda} \overline{\langle y, x \rangle} = \bar{\lambda} \langle x, y \rangle.$$

■

1.1.5. Theorem. — *Let V be a $\mathbb{C}$ -inner product space. Then for any $x, y \in V$ we have the following inequalities:*

$$1. |\langle x, y \rangle| \leq \|x\| \|y\|. \quad (\text{Cauchy-Schwarz inequality})$$

Equality holds $\iff x$ and y are linearly dependent, i.e. $x = \lambda y$ for some $\lambda \in \mathbb{C}$.

$$2. \|x + y\| \leq \|x\| + \|y\|. \quad (\text{Triangle inequality})$$

$$3. \|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2. \quad (\text{Parallelogram law})$$

Proof. 1. If $y = 0$ then this is trivial. So let $y \neq 0$. Then $\langle y, y \rangle \neq 0$. Now for any $\lambda \in \mathbb{C}$,

$$\begin{aligned} 0 &\leq \|x - \lambda y\|^2 \\ &= \langle x - \lambda y, x - \lambda y \rangle \\ &= \langle x, x - \lambda y \rangle - \lambda \langle y, x - \lambda y \rangle \\ &= \langle x, x \rangle - \bar{\lambda} \langle x, y \rangle - \lambda \langle y, x \rangle + \lambda \bar{\lambda} \langle y, y \rangle. \end{aligned}$$

Setting $\lambda = \frac{\langle x, y \rangle}{\langle y, y \rangle}$, we get

$$0 \leq \|x\|^2 - \frac{\langle x, y \rangle^2}{\|y\|^2}. \quad (1.1.7)$$

Equality holds in (1.1.7) iff $x = \lambda y$ for some $\lambda \in \mathbb{C}$.

2.

$$\begin{aligned} \|x + y\|^2 &= \langle x + y, x + y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\ &= \|x\|^2 + \langle x, y \rangle + \overline{\langle x, y \rangle} + \|y\|^2 \\ &= \|x\|^2 + 2 \operatorname{Re}(\langle x, y \rangle) + \|y\|^2 \\ &= \|x\|^2 + 2 \langle x, y \rangle + \|y\|^2 \\ &\leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 = (\|x\| + \|y\|)^2. \end{aligned}$$

3.

$$\begin{aligned} \|x + y\|^2 + \|x - y\|^2 &= \langle x + y, x + y \rangle + \langle x - y, x - y \rangle \\ &= 2 \langle x, x \rangle + 2 \langle y, y \rangle \\ &= 2\|x\|^2 + 2\|y\|^2. \end{aligned}$$

■

1.1.6. Theorem (Continuity of inner product). — If $\{x_n\}_n$ & $\{y_n\}_n$ are two sequences in an inner product space X , converging to x & y respectively, then $\langle x_n, y_n \rangle \rightarrow \langle x, y \rangle$ as $n \rightarrow \infty$.

Proof. Since $x_n \rightarrow x$ and $y_n \rightarrow y$ as $n \rightarrow \infty$, the sequences $\{x_n\}_n$ and $\{y_n\}_n$ are bounded, i.e. there exist $M_1, M_2 > 0$ such that

$$\|x_n\| < M_1 \quad \text{and} \quad \|y_n\| < M_2 \quad \forall n \in \mathbb{N}.$$

Now

$$\begin{aligned} |\langle x_n, y_n \rangle - \langle x, y \rangle| &= |\langle x_n, y_n \rangle - \langle x_n, y \rangle + \langle x_n, y \rangle - \langle x, y \rangle| \\ &\leq |\langle x_n, y_n \rangle - \langle x_n, y \rangle| + |\langle x_n, y \rangle - \langle x, y \rangle| \\ &= |\langle x_n, y_n - y \rangle| + |\langle x_n - x, y \rangle| \\ &\leq \|x_n\| \|y_n - y\| + \|x_n - x\| \|y\| \text{ (by Theorem 1.1.5)} \\ &< M_1 \|y_n - y\| + M_2 \|x_n - x\| \\ &\rightarrow 0 \quad \text{as } n \rightarrow \infty. \end{aligned}$$

Hence, $\langle x_n, y_n \rangle \rightarrow \langle x, y \rangle$ as $n \rightarrow \infty$.

■

1.1.7. Theorem (Polarization identity). — For any x, y in a complex inner product space,

$$\langle x, y \rangle = \frac{1}{4} \left\{ \|x + y\|^2 - \|x - y\|^2 + i\|x + iy\|^2 - i\|x - iy\|^2 \right\}.$$

Proof.

$$\begin{aligned} \text{RHS} &= \frac{1}{4} \left\{ \langle x + y, x + y \rangle - \langle x - y, x - y \rangle + i \langle x + iy, x + iy \rangle - i \langle x - iy, x - iy \rangle \right\} \\ &= \frac{1}{4} \left\{ \|x\|^2 + \langle x, y \rangle + \langle y, x \rangle + \|y\|^2 \right. \\ &\quad \left. - \|x\|^2 + \langle x, y \rangle + \langle y, x \rangle - \|y\|^2 \right. \\ &\quad \left. + i\|x\|^2 + \langle x, y \rangle - \langle y, x \rangle + i\|y\|^2 \right. \\ &\quad \left. - i\|x\|^2 + \langle x, y \rangle - \langle y, x \rangle - i\|y\|^2 \right\} \\ &= \langle x, y \rangle = \text{LHS}. \end{aligned}$$

1.1.8. Theorem. — $\|x\| = \sqrt{\langle x, x \rangle}$ satisfies all the conditions for a norm on an inner product space X over $\mathbb{C}$.

Proof. Clearly $\|x\| \geq 0$, and if $\|x\| = 0$ then $\langle x, x \rangle = 0 \implies x = 0$. So $\|x\| > 0$ and $\|x\| = 0$ iff $x = 0$.

To prove: $\|x + y\| \leq \|x\| + \|y\|$. We have

$$\|x + y\|^2 = \langle x + y, x + y \rangle = \|x\|^2 + \langle x, y \rangle + \langle y, x \rangle + \|y\|^2.$$

Using Theorem 1.1.5 we have $|\langle x, y \rangle| \leq \|x\|\|y\|$ and $|\langle y, x \rangle| = |\overline{\langle x, y \rangle}| \leq \|x\|\|y\|$. Also $\langle x, y \rangle + \langle y, x \rangle = \langle x, y \rangle + \overline{\langle x, y \rangle}$ is a real number, and

$$\langle x, y \rangle + \langle y, x \rangle \leq |\langle x, y \rangle| + |\langle y, x \rangle| \leq 2\|x\|\|y\|.$$

So we get

$$\|x + y\|^2 \leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 = (\|x\| + \|y\|)^2.$$

Hence, $\|x + y\| \leq \|x\| + \|y\|$.

Finally, if λ is a scalar in $\mathbb{C}$, then

$$\|\lambda x\| = \sqrt{\langle \lambda x, \lambda x \rangle} = \sqrt{\lambda \bar{\lambda} \langle x, x \rangle} = \sqrt{|\lambda|^2 \|x\|^2} = |\lambda| \|x\| \quad \forall \lambda \in \mathbb{C}.$$

Thus $\|x\| = \sqrt{\langle x, x \rangle}$ satisfies all the conditions of a norm. ■

1.1.9. Definition (Hilbert space). — A **Hilbert space** is a complete inner product space. (The notion was introduced by Hilbert in 1912, while studying integral equations.)

1.1.10. Example. — $\mathbb{R}^n$, ℓ^2 , and $\mathbb{C}^n$ are Hilbert spaces, while ℓ^p ($p \neq 2$) and $C[a, b]$ are not Hilbert spaces (Exercise 1.1.13 and Exercise 1.1.14 below).

1.1.11. Remark. — Every Hilbert space is a Banach space.

1.1.12. Theorem. — A Banach space is a Hilbert space if and only if the parallelogram law holds.

Proof. It is clear that all Hilbert spaces are Banach spaces where the parallelogram law holds (Theorem 1.1.5).

Now suppose that X is a Banach space where the parallelogram law holds. We have to prove X is a Hilbert space. We introduce an inner product in X by

$$\langle x, y \rangle = \frac{1}{4} \left\{ \|x + y\|^2 - \|x - y\|^2 \right\}. \quad (1.1.8)$$

Clearly $\langle x, x \rangle \geq 0$, $\langle x, x \rangle = 0$ iff $x = 0$. Also $\langle x, y \rangle = \langle y, x \rangle$.

To show $\langle x_1 + x_2, y \rangle = \langle x_1, y \rangle + \langle x_2, y \rangle$ and $\langle \lambda x, y \rangle = \lambda \langle x, y \rangle \quad \forall \lambda \in \mathbb{R}$.

By the parallelogram law we can write

$$\begin{aligned} \|u + v + w\|^2 + \|u + v - w\|^2 &= 2 \left(\|u + v\|^2 + \|w\|^2 \right), \\ \|u - v + w\|^2 + \|u - v - w\|^2 &= 2 \left(\|u - v\|^2 + \|w\|^2 \right). \end{aligned}$$

Subtracting we have

$$\begin{aligned} \|u + v + w\|^2 + \|u + v - w\|^2 - \|u - v + w\|^2 - \|u - v - w\|^2 &= 2 \left(\|u + v\|^2 - \|u - v\|^2 \right), \\ \frac{1}{4} \left(\|u + w + v\|^2 - \|u + w - v\|^2 \right) + \frac{1}{4} \left(\|u - w + v\|^2 - \|u - w - v\|^2 \right) \\ &= 2 \cdot \frac{1}{4} \left(\|u + v\|^2 - \|u - v\|^2 \right). \end{aligned}$$

By (1.1.8) we have

$$\langle u + w, v \rangle + \langle u - w, v \rangle = 2 \langle u, v \rangle.$$

Putting $w = u$ we obtain $\langle 2u, v \rangle = 2 \langle u, v \rangle$. Again putting $x_1 = u + w, x_2 = u - w$ & $x_3 = v$, we get

$$\langle x_1, x_3 \rangle + \langle x_2, x_3 \rangle = 2 \langle u, v \rangle = \langle 2u, v \rangle = \langle x_1 + x_2, x_3 \rangle,$$

so the condition $\langle x_1 + x_2, y \rangle = \langle x_1, y \rangle + \langle x_2, y \rangle$ is satisfied.

Now let $x_1, x_2, x_3 \in X$ (Banach space). $\langle x_1 + x_2 + x_3, x_4 \rangle = \langle x_1, x_4 \rangle + \langle x_2, x_4 \rangle + \langle x_3, x_4 \rangle$. Putting $x_1 = x_2 = x_3 = x$ and $x_4 = y$, we get $\langle 3x, y \rangle = 3 \langle x, y \rangle$. So, by the principle of mathematical induction, for any positive integer n

$$\langle nx, y \rangle = n \langle x, y \rangle \quad \forall x, y \in X.$$

Now

$$\langle -x, y \rangle = \frac{1}{4} \left\{ \| -x + y \|^2 - \| -x - y \|^2 \right\} = \frac{1}{4} \left\{ -\|x + y\|^2 + \|x - y\|^2 \right\} = -\langle x, y \rangle.$$

Taking $n = -m$, $m > 0$, m a positive integer, we have

$$\langle nx, y \rangle = \langle -mx, y \rangle = m \langle -x, y \rangle = -m \langle x, y \rangle = n \langle x, y \rangle.$$

So $\langle \lambda x, y \rangle = \lambda \langle x, y \rangle$ is satisfied for all positive and negative integers.

Now consider $\lambda = p/q$, $q \neq 0$, $\gcd(p, q) = 1$, p, q are integers. So $\langle \lambda x, y \rangle = \left\langle \frac{p}{q} x, y \right\rangle$. Then $q \langle \lambda x, y \rangle = q \left\langle \frac{p}{q} x, y \right\rangle = \langle px, y \rangle = p \langle x, y \rangle$, i.e. $\langle \lambda x, y \rangle = \frac{p}{q} \langle x, y \rangle$, i.e. $\langle \lambda x, y \rangle = \lambda \langle x, y \rangle$ where λ is rational.

Now let λ be any real. So there exists a sequence of rationals $\{r_n\}_n$ such that $r_n \rightarrow \lambda$ as $n \rightarrow \infty$.

$$|\langle r_n x, y \rangle - \langle \lambda x, y \rangle| = |\langle (r_n - \lambda)x, y \rangle| \leq \|(r_n - \lambda)x\| \|y\| \rightarrow 0 \text{ as } n \rightarrow \infty$$

(by Theorem 1.1.5). Thus, $\langle r_n x, y \rangle \rightarrow \langle \lambda x, y \rangle$ as $n \rightarrow \infty$, i.e. $r_n \langle x, y \rangle \rightarrow \langle \lambda x, y \rangle$ as $n \rightarrow \infty$, i.e. $\lambda \langle x, y \rangle = \langle \lambda x, y \rangle$.

Consequently, X is an inner product space, and as it is complete as a Banach space, X is a Hilbert space. ■

Note. — For the underlying space a complex linear space, we instead define

$$\langle x, y \rangle = \frac{1}{4} \left\{ \|x + y\|^2 - \|x - y\|^2 + i\|x + iy\|^2 - i\|x - iy\|^2 \right\} \quad \forall x, y \in X,$$

and proceeding as above we can show X is a Hilbert space over $\mathbb{C}$.

1.1.13. Exercise. — The space $C[a, b]$, the space of all real-valued continuous functions on $[a, b]$, with respect to the supnorm

$$\|x\| = \sup_{a \leq t \leq b} |x(t)|,$$

is not an inner product space, and hence not a Hilbert space.

Solution. We show that the norm defined by $\|x\| = \sup_{a \leq t \leq b} |x(t)|$ cannot be obtained from an inner product, since this norm doesn't satisfy the parallelogram equality. If we take

$$x(t) = 1, \quad y(t) = \frac{t-a}{b-a},$$

then $\|x\| = 1, \|y\| = 1$.

$$x(t) + y(t) = 1 + \frac{t-a}{b-a}, \quad x(t) - y(t) = 1 - \frac{t-a}{b-a} \implies \|x + y\| = 2, \|x - y\| = 1.$$

So $\|x + y\|^2 + \|x - y\|^2 = 5$, but $2(\|x\|^2 + \|y\|^2) = 4$. This completes the proof. ■

1.1.14. Exercise. — The space ℓ^p with $p \neq 2$ is not an inner product space, hence not a Hilbert space.

Solution. We show that the norm

$$\|x\| = \left(\sum_{i=1}^{\infty} |\xi_i|^p \right)^{1/p}, \quad \text{where } x = (\xi_1, \xi_2, \dots),$$

($p \neq 2$) cannot be obtained from an inner product. We prove the norm doesn't satisfy the parallelogram equality. Let us take

$$x = (1, 1, 0, 0, \dots) \in \ell^p, \quad y = (1, -1, 0, 0, \dots) \in \ell^p.$$

Then $\|x + y\| = 2 = \|x - y\|, \|x\| = \|y\| = 2^{1/p}$. But

$$\|x + y\|^2 + \|x - y\|^2 = 8 \neq 2(\|x\|^2 + \|y\|^2) = 2 \cdot 2 \cdot 2^{2/p} \quad \text{for all } p \neq 2.$$

So the parallelogram equality is not satisfied if $p \neq 2$.

ℓ^p is complete. Hence ℓ^p ($p \neq 2$) is a Banach space which is not a Hilbert space. ■

1.2 Orthogonality

1.2.1. Definition (Orthogonality). — Let L be a nonempty subset of H (Hilbert space). Two elements x and y of H are said to be **orthogonal** if their inner product $\langle x, y \rangle = 0$. We write $x \perp y$.

If x is orthogonal to every element of L , then we say that x is orthogonal to L , and we write $x \perp L$.

Two nonempty subsets F_1 and F_2 of a Hilbert space H are said to be orthogonal, written $F_1 \perp F_2$, if $x \perp y$ where $x \in F_1$ and $y \in F_2$.

1.2.2. Theorem (Pythagorean theorem). — If $x, y \in H$, and $x \perp y$, then

$$\|x + y\|^2 = \|x\|^2 + \|y\|^2.$$

Proof. Since $x \perp y$, so $\langle x, y \rangle = 0$. Then

$$\begin{aligned} \|x + y\|^2 &= \langle x + y, x + y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\ &= \|x\|^2 + \langle x, y \rangle + \overline{\langle x, y \rangle} + \|y\|^2 \\ &= \|x\|^2 + \|y\|^2. \end{aligned}$$

■

1.2.3. Definition (Orthogonal complement). — Let $S \subseteq H$. The set of all elements of H orthogonal to S is called the **orthogonal complement** of S , and is denoted by $S^\perp$. So

$$S^\perp = \{x \in H \mid \langle x, y \rangle = 0 \ \forall y \in S\}.$$

1.2.4. Theorem (Result). — The following hold:

1. $\{0\}^\perp = H$ and $H^\perp = \{0\}$.
2. If $S \subseteq H$, then $S \cap S^\perp = \{0\}$.
3. If $S_1 \subseteq S_2 \subseteq H$, then $S_2^\perp \subseteq S_1^\perp$.
4. If $S \subseteq H$, then $S^\perp$ is a closed subspace of H .

Proof. 1. Suppose $x \in H$. Then $\langle x, 0 \rangle = 0$, so $x \in \{0\}^\perp$. Thus $\{0\}^\perp = H$.

Suppose $x \in H^\perp$. Then $\langle x, y \rangle = 0 \ \forall y \in H$. Then $\langle x, x \rangle = 0$, so $x = 0$. Thus $H^\perp = \{0\}$.

2. Suppose $x \in S \cap S^\perp$. Then $\langle x, y \rangle = 0 \ \forall y \in S$, so $\langle x, x \rangle = 0$. Thus $x = 0$. Hence $S \cap S^\perp = \{0\}$.
3. Suppose $x \in S_2^\perp$. Then $\langle x, y \rangle = 0 \ \forall y \in S_2 \supseteq S_1$, hence $\langle x, y \rangle = 0 \ \forall y \in S_1 \subseteq S_2$. Hence $x \in S_1^\perp$. Thus $S_2^\perp \subseteq S_1^\perp$.
4. If $S \subseteq H$, then $S^\perp = \{x \in H \mid \langle x, y \rangle = 0 \ \forall y \in S\}$.

Let $\{x_n\}$ be a Cauchy sequence in $S^\perp$ such that $x_n \rightarrow x$ as $n \rightarrow \infty$. We have to show that $x \in S^\perp$.

Since $\{x_n\}_n \in S^\perp$, $\langle x_n, y \rangle = 0 \ \forall y \in S$.

$$\begin{aligned} \therefore \lim_{n \rightarrow \infty} \langle x_n, y \rangle &= 0 \quad \forall y \in S \implies \left\langle \lim_{n \rightarrow \infty} x_n, y \right\rangle = 0 \quad \forall y \in S \\ \implies \langle x, y \rangle &= 0 \quad \forall y \in S \implies x \in S^\perp. \end{aligned}$$

Then $S^\perp$ is a closed subspace of H . ■

1.2.5. Theorem. — *If M and N are two closed subspaces of a Hilbert space H such that $M \perp N$, then the subspace $(M + N)$ is also closed.*

Proof. Since $M \perp N$,

$$M \cap N = \{0\}.$$

So each element of $M + N$ can be expressed as $w = w_1 + w_2$ where $w_1 \in M$ & $w_2 \in N$.

Let $z_n \rightarrow z$, where $\{z_n\}_n \in M + N$. We have to show $z \in M + N$.

Let $z_n = x_n + y_n$, where $\{x_n\}_n \in M$ and $\{y_n\}_n \in N$.

Now $(x_m - x_n) \in M$ and $(y_m - y_n) \in N$ (as M & N are subspaces). Also $(x_m - x_n) \perp (y_m - y_n)$ as $M \perp N$.

Using the Pythagorean theorem (Theorem 1.2.2) we have

$$\begin{aligned} \|x_m - x_n\|^2 + \|y_m - y_n\|^2 &= \|(x_m - x_n) + (y_m - y_n)\|^2 \\ &= \|(x_m + y_m) - (x_n + y_n)\|^2 \\ &= \|z_m - z_n\|^2 \\ &\rightarrow 0 \quad \text{as } m, n \rightarrow \infty. \end{aligned}$$

It follows that $\{x_n\}_n$ and $\{y_n\}_n$ are both Cauchy sequences, in M and N respectively.

Since M and N are closed, they are complete (as H is Hilbert space). So there exist elements $x \in M$ and $y \in N$ such that $x_n \rightarrow x$ and $y_n \rightarrow y$ as $n \rightarrow \infty$.

Now $x + y \in M + N$, and as

$$\lim_{n \rightarrow \infty} z_n = \lim_{n \rightarrow \infty} (x_n + y_n) = \lim_{n \rightarrow \infty} x_n + \lim_{n \rightarrow \infty} y_n = x + y = z,$$

thus $(M + N)$ is a closed subspace of H . ■

1.2.6. Theorem (Projection theorem). — *If M is a closed subspace of H , then*

$$H = M \oplus M^\perp.$$

Proof. Since M is closed subspace of H , then $M^\perp$ is a closed subspace of H (Theorem 1.2.4). So $M \perp M^\perp$, and again by Theorem 1.2.5, $(M + M^\perp)$ is also closed subspace of H .

We want to show $H = M + M^\perp$ (as $M \cap M^\perp = \{0\}$). If this is not so, i.e. $H \neq (M + M^\perp)$, then there exists an element $z_0 \neq 0$, such that $z_0 \perp (M + M^\perp)$ (proof omitted).

So $\langle z_0, x + y \rangle = 0$ where $x \in M$ and $y \in M^\perp$.

Taking first $x = 0$, we have $\langle z_0, y \rangle = 0 \ \forall y \in M^\perp \implies z_0 \perp M^\perp \implies z_0 \in (M^\perp)^\perp = M$.

When $y = 0 \implies \langle z_0, x \rangle = 0 \forall x \in M \implies z_0 \perp M \implies z_0 \in M^\perp$.

Which implies $z_0 \in M \cap M^\perp$.

But we know $M \cap M^\perp = \{0\}$. Since $z_0 \neq 0$, it cannot be a member of $M \cap M^\perp$. This contradicts that $z_0 \in H \setminus (M + M^\perp)$.

$\therefore H = M + M^\perp$ & $M \cap M^\perp = \{0\}$ implies $H = M \oplus M^\perp$. ■

Note. — Here we have assumed $(M^\perp)^\perp = M$.

1.2.7. Theorem (Riesz representation theorem¹). — Let H be a Hilbert space. Then a function $f : H \rightarrow \mathbb{C}$ is a continuous linear functional on H if and only if there exists a unique element y in H such that

$$f(x) = \langle x, y \rangle \quad \forall x \in H, \quad \text{and} \quad \|f\| = \|y\|.$$

Proof. Sufficiency part.

Let y be a fixed element in a Hilbert space H , and we define a functional f by

$$f(x) = \langle x, y \rangle.$$

Clearly f is linear (as inner product is linear). By Schwartz's inequality,

$$|f(x)| = |\langle x, y \rangle| \leq \|x\| \|y\|,$$

so f is bounded and therefore f is continuous linear functional with

$$\|f\| \leq \|y\|. \tag{1.2.1}$$

Also if $y \neq 0$, then

$$\|f\| = \sup \{ |f(x)| : \|x\| = 1 \} \geq \left| f \left(\frac{y}{\|y\|} \right) \right| = \left| \left\langle \frac{y}{\|y\|}, y \right\rangle \right| = \|y\|.$$

$$\|f\| \geq \|y\|. \tag{1.2.2}$$

From (1.2.1) & (1.2.2), we have $\|f\| = \|y\|$.

If $y = 0$, it is obviously true.

Necessary part.

We first verify that if such y exists then it is unique.

If y' is not unique then let $y' \in H$ be such that we have $f(x) = \langle x, y' \rangle \quad \forall x \in H$.

Then $\langle x, y \rangle = \langle x, y' \rangle \quad \forall x \in H \implies \langle x, y - y' \rangle = 0 \quad \forall x \in H$. Putting $x = y - y'$, we get

$$\|y - y'\|^2 = \langle y - y', y - y' \rangle = 0 \implies y - y' = 0 \implies y = y'.$$

$\therefore y$ is unique.

Now if f is a zero functional, then we take $y = 0$ and then the equality $f(x) = \langle x, y \rangle$ is satisfied $\forall x \in H$.

Next, if f is a nonzero functional, and let M denote the set of elements $x \in H$ such that $f(x) = 0$. Then M is a non-void proper closed subspace of H . There exists a nonzero element y_0 such that $y_0 \perp M$.

¹Hahn-Banach theorem for Hilbert space.

We shall show that for a suitable scalar α ,

$$y = \alpha y_0.$$

We first show that whatever α may be, the equality $f(x) = \langle x, y \rangle$ holds for $\forall x \in M$. Because for such x , $f(x) = 0$, and as $y_0 \perp M$, we have

$$\langle x, \alpha y_0 \rangle = \alpha \langle x, y_0 \rangle = 0, \quad \forall x \in M.$$

Next we observe that after a suitable selection of α , the equality $f(x) = \langle x, y \rangle$ holds for $x = y_0$. For this, the condition on α is

$$f(y_0) = \langle y_0, y \rangle = \langle y_0, \alpha y_0 \rangle = \alpha \langle y_0, y_0 \rangle = \alpha \|y_0\|^2,$$

so we can select $\alpha = \frac{f(y_0)}{\|y_0\|^2}$.

Then $f(x) = \langle x, y \rangle$ is satisfied for $x = y$ and also for $x \in M$.

We also observe that $f(y_0) \neq 0$, because if $f(y_0) = 0$, then $y_0 \in M$, but $y_0 \perp M$, which will imply $y_0 = 0$, which is not the case here.

Let $x \in H$ and consider $x - \beta y_0$, where $\beta = \frac{f(x)}{f(y_0)}$. Then

$$f(x - \beta y_0) = f(x) - \beta f(y_0) = 0 \implies x - \beta y_0 \in M.$$

We consider $x - \beta y_0 = m$ (say). Thus $m \in M$, & every element $x \in H$ can be written as $x = m + \beta y_0$, where $m \in M$ and β is a suitable scalar.

$$\therefore f(x) = f(m + \beta y_0) = f(m) + \beta f(y_0) = \langle m, y \rangle + \beta \langle y_0, y \rangle = \langle m + \beta y_0, y \rangle = \langle x, y \rangle.$$

This proves the necessary part. ■

1.3 Orthonormality

1.3.1. Definition (Orthonormal set). — Let X be an inner product space and $M \subseteq X$. Then M is called **orthonormal** if $\forall x, y \in M$,

$$\langle x, y \rangle = \begin{cases} 0 & \text{if } x \neq y \\ 1 & \text{if } x = y. \end{cases}$$

1.3.2. Example. — In ℓ^2 -space, if $x = \{\xi_i\}$ and $y = \{\eta_i\} \forall x, y \in \ell^2$, and $\langle x, y \rangle = \sum_{i \geq 1} \xi_i \overline{\eta_i}$, then if

$$\begin{aligned} x_1 &= \{1, 0, 0, \dots\}, \\ x_2 &= \{0, 1, 0, \dots\}, \\ &\vdots \\ x_n &= \{\underbrace{0, 0, \dots, 0}_{n-1 \text{ places}}, 1, 0, 0, \dots\}, \end{aligned}$$

then

$$\langle x_i, x_j \rangle = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j. \end{cases}$$

So in ℓ^2 , $\{x_n\}_n$ is an orthonormal sequence.

1.3.3. Example. — Let X denote the inner product space of a real-valued continuous function on $[0, 2\pi]$, with inner product defined by

$$\langle x, y \rangle = \int_0^{2\pi} x(t)y(t) dt.$$

Then if $x_n(t) = \cos nt$,

$$\langle x_m, x_n \rangle = \int_0^{2\pi} \cos mt \cos nt dt = \begin{cases} 0 & \text{if } m \neq n \\ \pi & \text{if } m = n = 1, 2, \dots \\ 2\pi & \text{if } m = n = 0. \end{cases}$$

Now if $e_0(t) = \frac{1}{\sqrt{2\pi}}$, $e_n(t) = \frac{\cos nt}{\sqrt{\pi}}$, $n = 1, 2, \dots$, then we can easily verify that $\{e_n\}_n$ is an orthonormal sequence in $X = L^2[0, 2\pi]$.

1.3.4. Theorem. — *An orthonormal set in an inner product space is linearly independent.*

Proof. Suppose that $S = \{v_1, \dots, v_k\}$ and let $a_1, \dots, a_k$ be scalars such that

$$\sum_{i=1}^k a_i v_i = 0.$$

For $j = 1, \dots, k$, we have (by orthogonality)

$$\left\langle v_j, \sum_{i=1}^k a_i v_i \right\rangle = \sum_{i=1}^k a_i \langle v_j, v_i \rangle = a_j \langle v_j, v_j \rangle = a_j \|v_j\|^2.$$

Thus, applying an inner product with v_j to both sides of $\sum_{i=1}^k a_i v_i = 0$,

$$a_j \|v_j\|^2 = \left\langle v_j, \sum_{i=1}^k a_i v_i \right\rangle = \langle v_j, 0 \rangle = 0,$$

and, as $\|v_j\| = 1$ by orthonormality, this implies $a_j = 0$ for every j . Hence, S is linearly independent. ■

1.3.5. Theorem (Bessel's inequality). — *Let $\{e_1, e_2, \dots, e_n, \dots\}$ be an orthonormal sequence in a Hilbert space H , then for every $x \in H$*

$$\sum_{i=1}^{\infty} |\langle x, e_i \rangle|^2 \leq \|x\|^2.$$

Proof. Let n be a positive integer and $\xi_i = \langle x, e_i \rangle$. Then

$$\begin{aligned} 0 &\leq \left\langle x - \sum_{i=1}^n \xi_i e_i, x - \sum_{j=1}^n \xi_j e_j \right\rangle \\ &= \langle x, x \rangle - \left\langle x, \sum_{j=1}^n \xi_j e_j \right\rangle - \left\langle \sum_{i=1}^n \xi_i e_i, x \right\rangle + \left\langle \sum_{i=1}^n \xi_i e_i, \sum_{j=1}^n \xi_j e_j \right\rangle \end{aligned}$$

$$\begin{aligned}
&= \|x\|^2 - \sum_{j=1}^n \overline{\xi_j} \langle x, e_j \rangle - \sum_{i=1}^n \xi_i \langle e_i, x \rangle + \sum_{i=1}^n \sum_{j=1}^n \xi_i \overline{\xi_j} \langle e_i, e_j \rangle \\
&= \|x\|^2 - \sum_{j=1}^n \overline{\xi_j} \xi_j - \sum_{i=1}^n \xi_i \overline{\xi_i} + \sum_{i=1}^n |\xi_i|^2 \\
&0 \leq \|x\|^2 - \sum_{j=1}^n |\xi_j|^2 - \sum_{i=1}^n |\xi_i|^2 + \sum_{i=1}^n |\xi_i|^2.
\end{aligned}$$

$\therefore \sum_{j=1}^n |\xi_j|^2 \leq \|x\|^2$. Then $\sum_{j=1}^n |\langle x, e_j \rangle|^2 \leq \|x\|^2$. Since the relation is true for any positive integer n , so $\sum_{i=1}^{\infty} |\langle x, e_i \rangle|^2$ is convergent,

$$\& \sum_{i=1}^{\infty} |\langle x, e_i \rangle|^2 \leq \|x\|^2. \quad \blacksquare$$

1.3.6. Theorem. — If $\{e_i\}_i$ is an orthonormal sequence in a Hilbert space and $x \in H$, then

$$\left(x - \sum_{i=1}^{\infty} \langle x, e_i \rangle e_i \right) \text{ is orthogonal to } e_j \text{ for each } j.$$

Proof. Let $s = \sum_{i=1}^{\infty} \langle x, e_i \rangle e_i$ and $s_n = \sum_{i=1}^n \langle x, e_i \rangle e_i$. The infinite series $\sum_{i=1}^{\infty} \langle x, e_i \rangle e_i$ is convergent (following from Bessel's inequality, Theorem 1.3.5).

Then if $1 \leq j \leq n$,

$$\langle x, e_j \rangle - \langle s_n, e_j \rangle = \langle x, e_j \rangle - \left(\sum_{i=1}^n \langle x, e_i \rangle \langle e_i, e_j \rangle \right) = \langle x, e_j \rangle - \sum_{i=1}^n \langle x, e_i \rangle \langle e_i, e_j \rangle = \langle x, e_j \rangle - \langle x, e_j \rangle = 0.$$

Thus $\langle s_n, e_j \rangle = \langle x, e_j \rangle$.

Now using the continuity of inner product (Theorem 1.1.6) we have

$$\begin{aligned}
\left\langle x - \sum_{i=1}^{\infty} \langle x, e_i \rangle e_i, e_j \right\rangle &= \left\langle x - \lim_{n \rightarrow \infty} s_n, e_j \right\rangle \\
&= \langle x - s, e_j \rangle \\
&= \langle x, e_j \rangle - \langle s, e_j \rangle \\
&= \langle x, e_j \rangle - \lim_{n \rightarrow \infty} \langle s_n, e_j \rangle \\
&= \langle x, e_j \rangle - \langle x, e_j \rangle = 0.
\end{aligned}$$

$\therefore (x - \sum_{i=1}^{\infty} \langle x, e_i \rangle e_i)$ is orthogonal to e_j . ■

1.3.7. Definition (Complete orthonormal set). — Let H be a Hilbert space (or inner product space). An orthonormal set $M \subseteq H$ is called a **complete orthonormal set** if there exists no orthonormal set of which M is a proper subset.

1.3.8. Example. — A countable orthonormal set is always a complete orthonormal set.

1.3.9. Theorem. — Let H be a Hilbert space and let $\{e_i\}_i$ be an orthonormal sequence in H . Then the following conditions are equivalent:

1. $\{e_i\}$ is complete.
2. $x \perp e_i \quad \forall i = 1, 2, \dots \implies x = 0$.
3. For every $x \in H$, x can be expressed as²

$$x = \sum_{i=1}^{\infty} \langle x, e_i \rangle e_i.$$

4. For every $x \in H$,

$$\|x\|^2 = \sum_{i=1}^{\infty} |\langle x, e_i \rangle|^2.$$

Proof. (i) $\implies$ (ii). Suppose (i) is true but (ii) is not true. Then there exists an element x ($\neq 0$), such that $x \perp e_i$, for $i = 1, 2, \dots$.

If $e = \frac{x}{\|x\|}$, then $\|e\| = 1$, and $e \perp e_i$ as $x \perp e_i$, for $i = 1, 2, \dots$.

So $\{e, e_1, e_2, \dots\}$ is an orthonormal set that contains properly the orthonormal set $\{e_1, e_2, \dots\}$ which contradicts that $\{e_i\}_i$ is complete.

Hence, (ii) is true.

Now we prove (ii) $\implies$ (iii). By previous theorem (Theorem 1.3.6), $(x - \sum_{i=1}^{\infty} \langle x, e_i \rangle e_i)$ is orthogonal to e_j , for each j .

Then by (ii), $x - \sum_{i=1}^{\infty} \langle x, e_i \rangle e_i = 0$.

Thus, for each $x \in H$,

$$x = \sum_{i=1}^{\infty} \langle x, e_i \rangle e_i.$$

For (iii) $\implies$ (iv), find $\|x\|^2 = \langle x, x \rangle = \langle \sum_{i=1}^{\infty} \langle x, e_i \rangle e_i, \sum_{i=1}^{\infty} \langle x, e_i \rangle e_i \rangle$.

Now (iv) $\implies$ (i). Let (iv) is true but (i) is not true.

Then there exists an orthonormal set $\{e, e_1, e_2, \dots\}$ where $\{e_i\}_i$ is a proper subset, where $\|e\| = 1$ & $e \perp e_i$,

$$\text{i.e.} \quad \langle e, e_i \rangle = 0, \quad \text{for } i = 1, 2, \dots$$

Since (iv) holds, $\|e\|^2 = \sum_{i=1}^{\infty} |\langle e, e_i \rangle|^2 = 0$, which shows that $e = 0$, if it contradicts that $\|e\| = 1$.

Thus, $\{e_i\}_i$ is complete. Hence, the theorem is proved. ■

Note. — 1. $\|x\|^2 = \sum_{i=1}^{\infty} |\langle x, e_i \rangle|^2$ is called **Parseval's equation**.

2. $x = \sum_{i=1}^{\infty} \langle x, e_i \rangle e_i$, $\langle x, e_i \rangle$ is called the **Fourier coefficient** of the expansion, is called the **Fourier expansion** of x .

1.3.10. Theorem. — If Parseval's equation holds for $x \in H$, then for every pair of elements $x, y \in H$, the **general Parseval's equation**

$$\langle x, y \rangle = \sum_{k=1}^{\infty} \langle x, e_k \rangle \langle e_k, y \rangle \quad \text{holds.}$$

²Fourier expansion in $L^2[0, 2\pi]$.

Proof. If λ is a scalar, then from Parseval's equation

$$\|x + \lambda y\|^2 = \sum_{k=1}^{\infty} |\langle x + \lambda y, e_k \rangle|^2 \quad \text{will hold,}$$

$$\begin{aligned} \text{i.e., } & \langle x, x \rangle + \lambda \langle y, x \rangle + \bar{\lambda} \langle x, y \rangle + |\lambda|^2 \langle y, y \rangle \\ &= \sum_{k=1}^{\infty} \left\{ \langle x + \lambda y, e_k \rangle \overline{\langle x + \lambda y, e_k \rangle} \right\} \end{aligned}$$

$$\begin{aligned} \text{i.e., } & \|x\|^2 + \lambda \langle y, x \rangle + \bar{\lambda} \langle x, y \rangle + |\lambda|^2 \|y\|^2 \\ &= \sum_{k=1}^{\infty} \left\{ |\langle x, e_k \rangle|^2 + \lambda \langle y, e_k \rangle \langle e_k, x \rangle \right. \\ & \quad \left. + \bar{\lambda} \langle x, e_k \rangle \langle e_k, y \rangle + |\lambda|^2 |\langle y, e_k \rangle|^2 \right\}. \end{aligned}$$

$$\text{As } \|x\|^2 = \sum_{k=1}^{\infty} |\langle x, e_k \rangle|^2 \text{ \& } \|y\|^2 = \sum_{k=1}^{\infty} |\langle y, e_k \rangle|^2,$$

$$\lambda \langle y, x \rangle + \bar{\lambda} \langle x, y \rangle = \lambda \sum_{k=1}^{\infty} \langle y, e_k \rangle \langle e_k, x \rangle + \bar{\lambda} \sum_{k=1}^{\infty} \langle x, e_k \rangle \langle e_k, y \rangle. \quad (1.3.1)$$

The relation (1.3.1) holds for arbitrary complex λ . Putting $\lambda = 1$ in (1.3.1), we get

$$\langle y, x \rangle + \langle x, y \rangle = \sum_{k=1}^{\infty} \langle y, e_k \rangle \langle e_k, x \rangle + \sum_{k=1}^{\infty} \langle x, e_k \rangle \langle e_k, y \rangle, \quad (1.3.2)$$

and again, putting $\lambda = i$ in (1.3.1), we get

$$\langle y, x \rangle - \langle x, y \rangle = \sum_{k=1}^{\infty} \langle y, e_k \rangle \langle e_k, x \rangle - \sum_{k=1}^{\infty} \langle x, e_k \rangle \langle e_k, y \rangle. \quad (1.3.3)$$

Subtracting (1.3.3) from (1.3.2), we get

$$\langle x, y \rangle = \sum_{k=1}^{\infty} \langle x, e_k \rangle \langle e_k, y \rangle. \quad \blacksquare$$

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